

“Open Source GIS Approach for Identifying Optimal Solid Waste Management Sites in Gorakhpur”

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submitted**

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by

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DECLARATION

I, Adarsh Pal, bearing 2205250000003 of Seventh Semester B. Tech, Department of Civil Engineering, Buddha Institute of Technology, GIDA, Gorakhpur declare that the Major Project entitled “**Open Source GIS Approach for Identifying Optimal Solid Waste Management Sites in Gorakhpur**” has been carried out by us and submitted in partial fulfillment of the course requirements for the award of degree in Bachelor of Technology in Civil Engineering from Dr. APJ Abdul Kalam Technical University, Lucknow during the academic year 2024-2025.

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ABSTRACT

Rapid urbanization and population growth in Indian cities have led to a significant increase in municipal solid waste (MSW) generation, posing serious environmental and management challenges. Gorakhpur, a major urban center in eastern Uttar Pradesh, faces critical issues related to unscientific waste disposal, shallow groundwater vulnerability, and flood-prone low-lying terrain. These conditions make landfill site selection a complex yet essential task for sustainable waste management. The present study aims to identify suitable landfill sites in Gorakhpur City using an open-source Geographic Information System (GIS)-based Multi-Criteria Evaluation (MCE) approach integrated with the Analytical Hierarchy Process. Eight key spatial criteria were considered, including land elevation, land use/land cover (LULC), proximity to roads, urban settlements, surface water bodies, railway networks, groundwater depth, and sensitive locations. Each criterion was processed, standardized, and assigned weights based on its relative importance in landfill suitability assessment. A weighted overlay analysis was performed using open-source GIS platforms such as QGIS, GRASS GIS, and SAGA GIS to generate a comprehensive landfill site suitability map. The results indicate that a large portion of the study area is unsuitable for landfill development due to dense urbanization, shallow groundwater levels, and environmental constraints. However, approximately 12–18% of the total area, primarily located in the western, northwestern, and southwestern peripheral regions, was identified as suitable. These areas exhibit favorable conditions such as deeper groundwater levels, reduced environmental sensitivity, and lower population density. The study demonstrates that GIS-based MCE integrated with AHP provides an effective and reliable framework for landfill site selection. The findings can support urban planners and municipal authorities in making informed, data-driven decisions for sustainable solid waste management in Gorakhpur City.

Keywords: MSW, GIS-Based Site Selection, AHP, Landfill Suitability Mapping

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1.1 Background

Rapid urbanization, population growth, and changes in consumption patterns have led to a significant increase in municipal solid waste (MSW) generation worldwide. Managing this waste effectively has become a major environmental and public health challenge, particularly in developing countries where urban infrastructure often struggles to keep pace with increasing waste volumes (Tchobanoglous et al., 1993; Hoornweg & Bhada-Tata, 2012).

India is currently experiencing rapid urban expansion, which has resulted in a substantial increase in municipal solid waste generation. According to the Central Pollution Control Board, Indian urban areas generate approximately 160,000–180,000 tonnes of municipal solid waste per day, and this quantity continues to increase due to rapid population growth and changing lifestyles (CPCB, 2021). Efficient solid waste management requires a systematic approach involving waste collection, transportation, processing, recycling, and safe disposal.

Among these stages, the final disposal of solid waste through landfill systems remains one of the most critical components of urban waste management. Improper landfill siting can lead to severe environmental consequences such as groundwater contamination, soil pollution, air pollution from methane emissions, and adverse impacts on human health (Kontos et al., 2005; Tchobanoglous et al., 1993).

In many developing countries, including India, municipal waste is often disposed of in open dumps without proper environmental safeguards. Such practices allow the formation of leachate, a liquid produced when water percolates through waste materials and dissolves hazardous substances. Leachate infiltration into soil layers can contaminate groundwater resources, posing serious risks to drinking water supplies and ecosystem health (Gupta et al., 2019).

With increasing environmental concerns and stricter environmental regulations, the need for scientifically selected landfill sites has become more important than ever before.

1.2 Importance of Scientific Landfill Site Selection

Landfill site selection is a complex spatial decision-making process that requires the evaluation of multiple environmental, social, and technical parameters simultaneously. The suitability of a landfill site depends on several factors including land use patterns, distance from residential areas, proximity to water bodies, groundwater depth, transportation accessibility, and topographical characteristics (Kontos et al., 2005; Nas et al., 2010).

Selecting inappropriate landfill sites can result in long-term environmental degradation and economic losses. Therefore, careful evaluation of spatial criteria is essential to ensure environmentally sustainable waste management. A scientifically selected landfill site provides several benefits, including:

- i. Protection of groundwater and surface water resources
- ii. Reduction of environmental pollution
- iii. Lower transportation and operational costs
- iv. Protection of public health and nearby communities
- v. Long-term sustainability of waste disposal systems

Traditional landfill siting approaches often relied on administrative decisions or land availability without considering environmental impacts. However, modern spatial technologies allow for a more systematic and objective evaluation of landfill suitability.

1.3 Role of GIS in Landfill Site Selection

Geographic Information Systems (GIS) have become one of the most powerful tools for environmental planning and spatial decision-making. GIS enables the integration, analysis, and visualization of multiple spatial datasets, making it particularly suitable for multi-criteria decision analysis in landfill site selection (Malczewski, 2006).

GIS-based landfill suitability analysis involves the integration of different thematic layers such as land use and land cover, elevation, groundwater depth, road networks, settlements, water bodies, and environmentally sensitive areas. These layers are analyzed through spatial techniques such as buffer analysis, overlay analysis, and weighted multi-criteria evaluation to identify potential landfill locations (Nas et al., 2010).

Recent developments in open-source GIS platforms have significantly improved access to spatial analysis tools. Software such as QGIS, GRASS GIS, and SAGA GIS provide

advanced geospatial analysis capabilities without the need for expensive proprietary software. Open-source GIS tools offer several advantages including:

- a. Cost-effectiveness
- b. Transparency in analytical procedures
- c. Reproducibility of research results
- d. Accessibility for academic institutions and government agencies

These features make open-source GIS platforms highly suitable for environmental planning and waste management studies.

1.4 Solid Waste Management Challenges in Gorakhpur

Gorakhpur is an important urban center in eastern Uttar Pradesh and has experienced rapid urban growth over the past few decades. The expansion of urban settlements and population growth have significantly increased the generation of municipal solid waste within the city. The geographical characteristics of Gorakhpur further complicate waste management challenges. The city is located in the Terai region of the Indo-Gangetic plains, which is characterized by flat topography, fertile alluvial soils, and a dense river network. These environmental conditions make the region highly vulnerable to flooding and groundwater contamination. Several factors contribute to waste management challenges in Gorakhpur:

- a. Rapid and unplanned urban expansion
- b. High population density
- c. Flood-prone terrain
- d. Shallow groundwater levels
- e. Limited availability of suitable land for landfill development
- f. Presence of ecologically sensitive wetlands

Improper waste disposal in such environmentally sensitive regions can cause severe environmental degradation and increase the risk of groundwater contamination (Gupta et al., 2019). Therefore, identifying environmentally suitable landfill sites using scientific spatial analysis methods is essential for sustainable waste management in Gorakhpur.

1.5 Research Gap

Although several studies have applied GIS-based techniques for landfill site selection, most of these studies focus on large metropolitan areas or rely on proprietary GIS software. Limited research has been conducted using open-source GIS tools for medium-sized cities such as Gorakhpur.

In addition, many previous studies have not adequately incorporated important environmental parameters such as groundwater depth, proximity to sensitive locations, and detailed land use patterns in landfill suitability analysis. The major research gaps identified include:

- a. Limited application of open-source GIS tools in landfill site selection studies
- b. Insufficient integration of hydrological parameters such as groundwater depth
- c. Lack of spatial analysis tailored to medium-sized cities
- d. Limited consideration of environmentally sensitive locations
- e. Lack of high-resolution spatial analysis for local planning

This study attempts to address these gaps by applying a comprehensive GIS-based multi-criteria evaluation using open-source geospatial tools to identify optimal waste disposal sites in Gorakhpur City.

1.6 Aim of the Study

The primary aim of this research is to identify optimal solid waste disposal sites in Gorakhpur City using a multi-criteria decision analysis approach based on open-source GIS techniques.

1.7 Objectives of the Study

The specific objectives of the study are:

- i. To collect and integrate spatial datasets influencing landfill suitability, including DEM, land use/land cover, groundwater depth, and transportation networks.
- ii. To generate thematic layers such as distance from roads, settlements, railways, water bodies, and environmentally sensitive locations.
- iii. To reclassify and assign suitability weights to each criterion based on environmental considerations.
- iv. To perform weighted overlay analysis using GIS to generate landfill suitability zones.

- v. To identify and recommend environmentally suitable sites for solid waste disposal within Gorakhpur City.

1.8 Study Area

The present study focuses on Gorakhpur City, an important urban center located in the eastern part of the state of Uttar Pradesh, India. Geographically, the city lies between 26°40'N to 26°50'N latitude and 83°20'E to 83°32'E longitude, within the fertile plains of the Indo-Gangetic basin. The region is characterized by relatively flat topography, alluvial soil deposits, and a dense network of rivers and wetlands, which significantly influence land use patterns and environmental conditions.

Gorakhpur is situated in the Terai region, which forms the transition zone between the Himalayan foothills and the Gangetic plains. The region experiences a humid subtropical climate with hot summers, a pronounced monsoon season, and mild winters. Annual rainfall is largely concentrated during the monsoon months, often leading to waterlogging and urban flooding in low-lying areas.

The rapid population growth and urban expansion in Gorakhpur have significantly increased municipal solid waste generation, placing additional pressure on the city's waste management infrastructure. The presence of shallow groundwater levels and nearby wetlands further increases the environmental risks associated with improper waste disposal. Consequently, identifying environmentally suitable locations for solid waste disposal is essential to minimize ecological impacts and protect groundwater resources.

The geographical location and administrative boundaries of the study area are illustrated in Figure 1, which shows the spatial position of Gorakhpur City within Uttar Pradesh, Gorakhpur District, and Gorakhpur Tehsil. The figure highlights the hierarchical administrative structure and the spatial extent of the study area used for the GIS-based landfill suitability analysis.

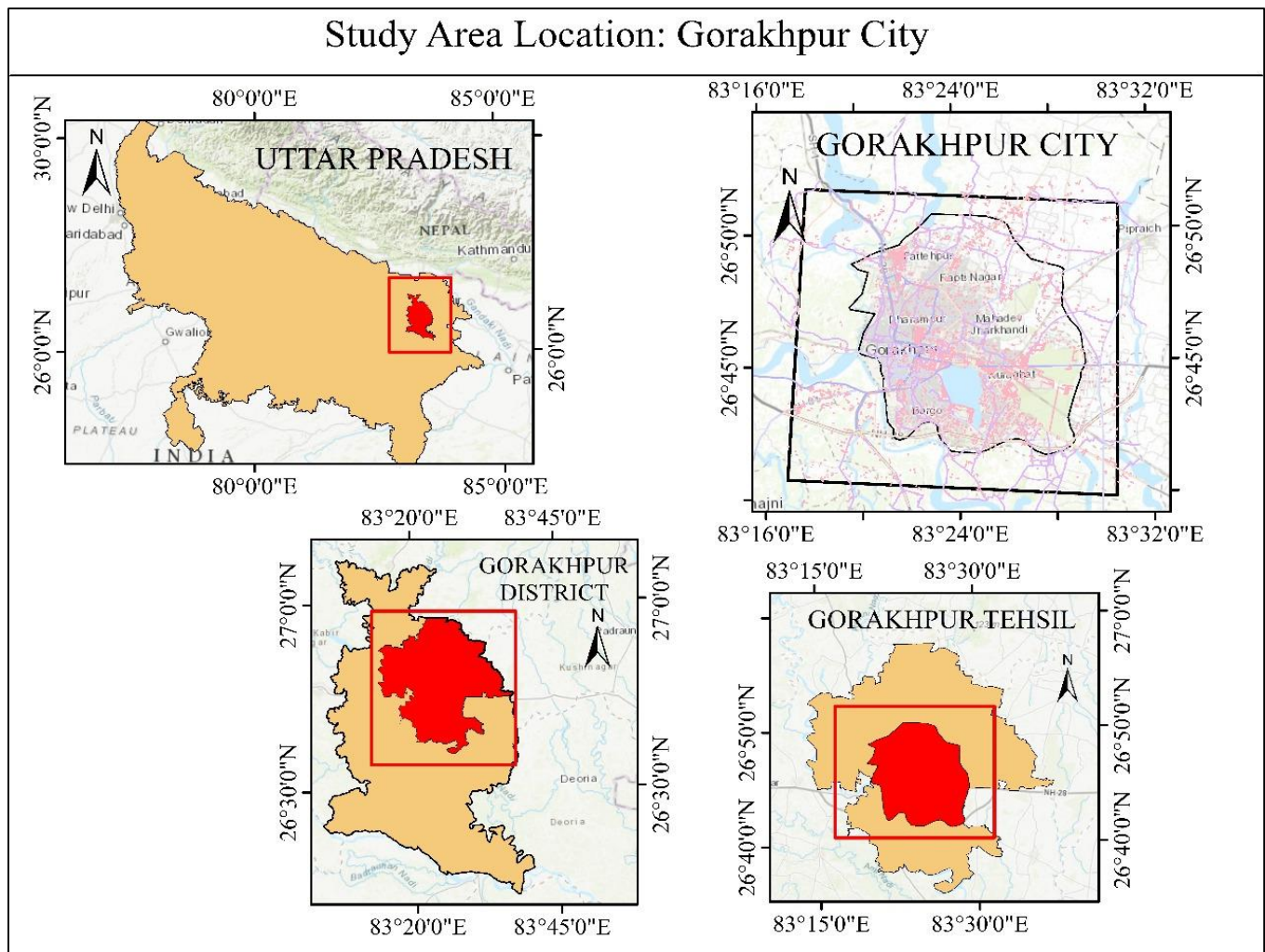


Figure 1. Study area location map showing the geographical position of Gorakhpur City within Uttar Pradesh, India. The figure illustrates the hierarchical administrative context including Uttar Pradesh State, Gorakhpur District, Gorakhpur Tehsil, and the spatial boundary of the study area used for the GIS-based solid waste landfill site suitability analysis.

2.1 Overview of Solid Waste Management

Solid waste management (SWM) is a critical component of sustainable urban development and environmental protection. Rapid urbanization, industrialization, and population growth have significantly increased the quantity and complexity of municipal solid waste generated worldwide (Hoornweg & Bhada-Tata, 2012). Municipal solid waste typically includes household waste, commercial waste, institutional waste, construction debris, and street sweepings.

Effective solid waste management involves a series of interconnected stages including waste generation, segregation, storage, collection, transportation, treatment, recycling, and final disposal (Tchobanoglous et al., 1993). Among these stages, the final disposal of waste through sanitary landfills remains one of the most widely adopted methods globally, particularly in developing countries where advanced waste treatment technologies are limited (Gupta et al., 2015).

Improper solid waste disposal can lead to severe environmental and public health problems. Open dumping and poorly designed landfill sites can result in groundwater contamination through leachate infiltration, air pollution due to methane emissions, soil degradation, and the spread of vector-borne diseases (Kontos et al., 2005). According to the Central Pollution Control Board, Indian cities collectively generate more than 160,000 tonnes of municipal solid waste per day, much of which is disposed of in open dumping sites lacking environmental safeguards (CPCB, 2021).

Therefore, scientifically designed landfill sites and efficient waste management systems are essential for minimizing environmental impacts and protecting public health.

2.2 GIS-based site suitability methods

Selecting appropriate landfill sites is one of the most important tasks in solid waste management planning. The suitability of a landfill location depends on several environmental, social, and technical criteria such as proximity to residential areas, distance

from water bodies, groundwater depth, road accessibility, land use patterns, and topography (Nas et al., 2010).

Traditional landfill site selection methods often relied on manual surveys and administrative decisions, which frequently overlooked environmental considerations and spatial relationships between different factors (Kontos et al., 2005). As a result, many landfill sites were located in environmentally sensitive areas, leading to long-term ecological damage.

Modern landfill siting methods increasingly rely on spatial analysis and decision-support systems to evaluate multiple criteria simultaneously. These methods include Geographic Information Systems (GIS), remote sensing, and multi-criteria decision analysis (MCDA) techniques (Malczewski, 2006). Several environmental criteria are commonly considered in landfill site selection studies, including:

- a) Distance from surface water bodies
- b) Distance from urban settlements
- c) Distance from transportation networks
- d) Land use and land cover
- e) Groundwater depth
- f) Slope and elevation
- g) Proximity to environmentally sensitive locations

Considering these criteria together helps planners identify landfill sites that minimize environmental risks and transportation costs while ensuring long-term sustainability.

2.3 GIS Applications in Waste Management

Geographic Information Systems (GIS) have become an essential tool for spatial data management and environmental planning. GIS enables researchers and planners to collect, analyze, visualize, and interpret spatial information related to environmental conditions and infrastructure networks (Longley et al., 2015). In the context of waste management, GIS plays a crucial role in:

- a) Waste collection route optimization
- b) Waste transportation planning
- c) Landfill site suitability analysis
- d) Environmental impact assessment

- e) Monitoring and management of landfill facilities

GIS allows the integration of multiple thematic datasets such as land use maps, digital elevation models (DEM), road networks, settlement locations, and hydrological features. These datasets can be analyzed simultaneously to identify suitable landfill locations (Nas et al., 2010).

Recent advancements in geospatial technology have also promoted the use of open-source GIS software for environmental analysis. Platforms such as QGIS, GRASS GIS, and SAGA GIS provide powerful spatial analysis tools without requiring expensive proprietary software licenses. These platforms enable researchers to perform advanced spatial analysis including buffer analysis, overlay analysis, raster reclassification, and suitability modeling.

Because of these capabilities, GIS has become one of the most widely used tools for landfill site selection and environmental planning studies.

2.4 Multi-Criteria Decision Analysis (MCDA) in GIS

Multi-Criteria Decision Analysis (MCDA) is a widely used approach for solving complex decision-making problems involving multiple conflicting criteria. In landfill site selection, MCDA helps evaluate various environmental and socio-economic factors simultaneously to determine the most suitable locations for waste disposal facilities (Malczewski, 2006).

MCDA methods are commonly integrated with GIS to perform spatial decision analysis. In a GIS-MCDA framework, different spatial layers representing environmental criteria are assigned weights based on their relative importance. These weighted layers are then combined through overlay analysis to generate a suitability map (Saaty, 1980). Several MCDA techniques have been applied in landfill site selection studies, including:

- a) Analytical Hierarchy Process (AHP)
- b) Weighted Linear Combination (WLC)
- c) Fuzzy logic models
- d) Boolean overlay methods

Among these methods, the Analytical Hierarchy Process (AHP) is one of the most commonly used techniques for assigning weights to different criteria based on pairwise comparisons (Saaty, 1980).

GIS-based MCDA provides a systematic framework for evaluating spatial criteria and helps

reduce subjectivity in decision-making processes. As a result, it has become a widely accepted method for environmental planning and waste management studies.

2.5 Review of Previous Studies on Landfill Site Selection

Over the past few decades, numerous researchers have applied Geographic Information Systems (GIS) and Multi-Criteria Decision Analysis (MCDA) techniques to identify suitable landfill sites in different geographical regions. These studies highlight the effectiveness of integrating spatial analysis with decision-making frameworks for environmentally sustainable waste management.

One of the early and significant contributions in this field was made by Kontos et al. (2005), who developed a GIS-based multi-criteria methodology for landfill site selection in Greece. Their study incorporated environmental, economic, and social parameters such as distance from settlements, water bodies, road networks, and geological characteristics. By applying weighted overlay techniques, the study demonstrated that GIS-based MCDA significantly improves the objectivity and transparency of landfill siting decisions compared to traditional approaches.

Similarly, Nas et al. (2010) conducted a comprehensive GIS-based multi-criteria evaluation for landfill site selection in Konya, Turkey. Their methodology involved the integration of multiple spatial datasets including land use/land cover, transportation networks, hydrological features, and slope. The study applied weighted linear combination techniques to generate suitability maps, emphasizing that GIS-based approaches can effectively support urban planning and environmental protection.

In the Indian context, Gupta et al. (2015) provided an extensive review of solid waste management practices and highlighted the urgent need for scientific landfill site selection methods. Their study emphasized that improper waste disposal practices in India have led to serious environmental issues such as groundwater contamination and air pollution. The authors recommended the use of GIS-based approaches for identifying environmentally suitable landfill locations and improving waste management systems.

Further advancements in landfill siting methodologies have incorporated the integration of remote sensing data with GIS techniques. According to Longley et al. (2015), remote sensing provides high-resolution spatial data on land use patterns, vegetation cover, and

topographical variations, which enhances the accuracy and reliability of GIS-based spatial analysis. The integration of remote sensing with GIS enables better monitoring and assessment of environmental conditions, making it highly suitable for landfill site selection studies.

Several researchers have also emphasized the importance of multi-criteria decision-making frameworks such as the Analytical Hierarchy Process (AHP) in landfill site selection. These approaches allow the assignment of weights to different criteria based on their relative importance, thereby improving the accuracy of suitability analysis (Malczewski, 2006; Saaty, 1980). The combination of GIS with AHP has been widely adopted in recent studies to reduce subjectivity and improve decision-making efficiency.

More recent studies have focused on incorporating additional environmental parameters such as groundwater vulnerability, proximity to sensitive locations, and urban expansion patterns into landfill suitability analysis. These studies highlight the importance of considering both environmental and socio-economic factors to ensure sustainable landfill site selection.

Overall, the reviewed literature clearly indicates that GIS-based multi-criteria analysis is a highly effective and reliable approach for landfill site selection. The integration of spatial datasets, decision-making models, and environmental criteria enables researchers and planners to identify suitable landfill locations while minimizing environmental risks and ensuring long-term sustainability.

2.6 Research Gap Identified

Although several studies have explored GIS-based landfill site selection techniques, significant research gaps remain. Many existing studies focus primarily on large metropolitan cities, while medium-sized cities often receive limited attention in scientific research.

Additionally, many landfill siting studies rely on proprietary GIS software, which may limit accessibility for researchers and local governments with limited financial resources. The use of open-source GIS platforms in landfill site selection studies remains relatively limited.

Another important gap in previous research is the insufficient integration of hydrological parameters such as groundwater depth, which plays a crucial role in preventing groundwater contamination from landfill leachate.

Furthermore, many studies do not adequately consider the influence of sensitive locations

such as educational institutions, hospitals, and protected areas on landfill suitability.

The present study addresses these gaps by applying an open-source GIS-based multi-criteria evaluation approach to identify suitable landfill locations in Gorakhpur City. The study integrates multiple environmental and infrastructural criteria including groundwater depth, land use patterns, transportation networks, and proximity to sensitive locations to generate landfill suitability zones.

3.1 Overview of Methodology

The present study adopts a Geographic Information System (GIS)-based Multi-Criteria Decision Analysis (MCDA) approach to identify optimal solid waste disposal sites in Gorakhpur City. GIS provides a powerful platform for integrating, analyzing, and visualizing spatial datasets, while MCDA enables the evaluation of multiple environmental, socio-economic, and infrastructural criteria simultaneously (Malczewski, 2006; Saaty, 1980).

The integration of GIS and MCDA has been widely recognized as an effective tool for landfill site selection, as it allows systematic assessment of various spatial parameters such as land use, proximity to water bodies, transportation networks, groundwater conditions, and environmentally sensitive areas (Kontos et al., 2005; Nas et al., 2010).

In this study, multiple thematic layers representing environmental and infrastructural factors were generated and analyzed using open-source GIS tools. These layers were standardized, reclassified, and assigned weights based on their relative importance in landfill site selection. The final suitability map was produced using a weighted overlay analysis, which combines all criteria into a single decision-support output (Malczewski, 2006). The overall workflow of the study includes the following steps:

- a) Data collection and preparation
- b) Data preprocessing and standardization
- c) Generation of thematic layers
- d) Buffer analysis and distance mapping
- e) Reclassification of criteria
- f) Assignment of weights using MCDA techniques
- g) Weighted overlay analysis
- h) Identification of suitable landfill zones

The detailed methodological workflow adopted in this study is illustrated in Figure 2, which presents the step-by-step process of data integration, spatial analysis, and suitability modeling for landfill site selection.

This methodology ensures a systematic, objective, and reproducible framework for landfill site selection. The use of open-source GIS platforms enhances transparency, cost-effectiveness, and accessibility, making the approach suitable for both academic research and real-world urban planning applications (Longley et al., 2015).

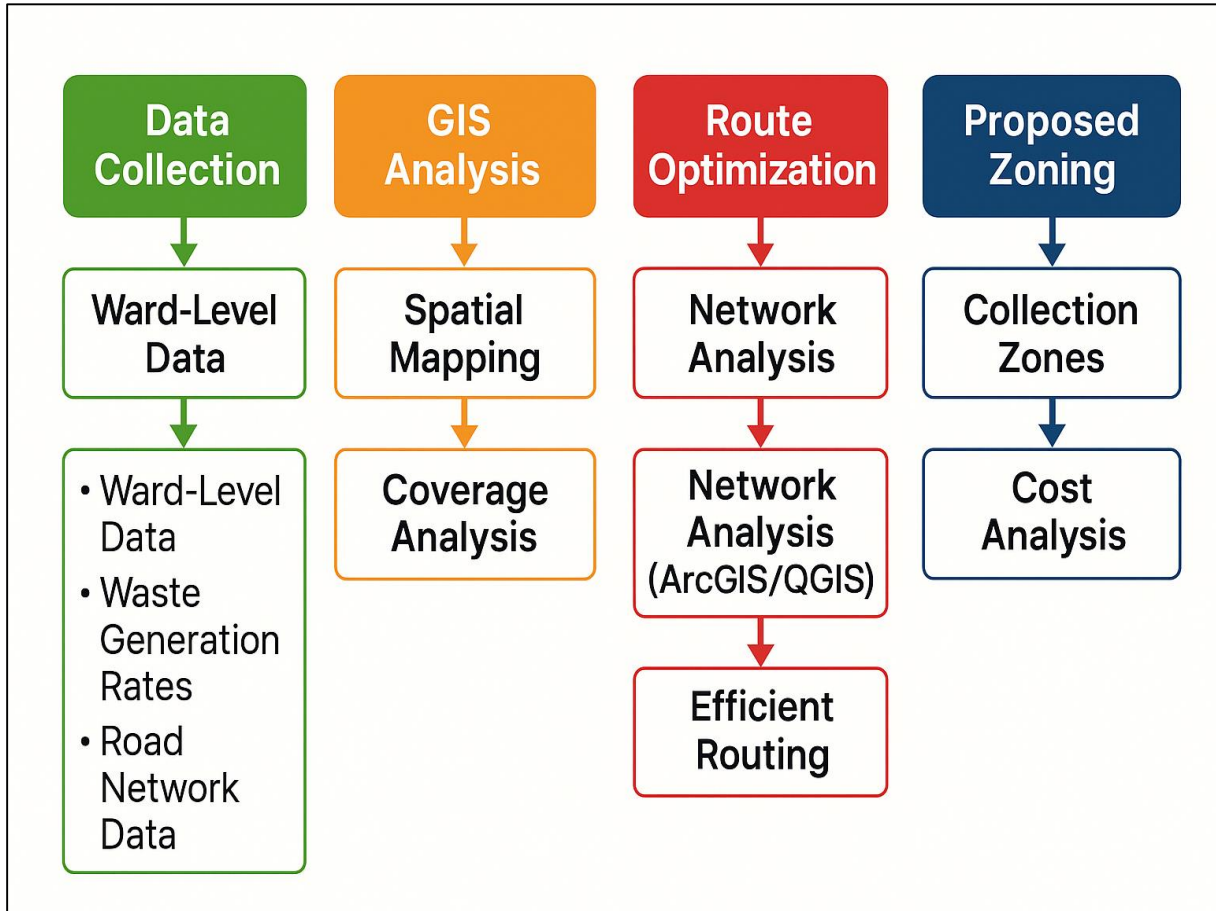


Figure 2. Methodological framework showing the GIS-based multi-criteria decision analysis (MCDA) approach for landfill site suitability assessment, including data preparation, spatial analysis, reclassification, weighting, and suitability mapping.

3.2 Data Used

A variety of spatial datasets were collected from open-source platforms and government databases to support the GIS-based multi-criteria analysis for landfill site selection. These datasets represent key environmental, infrastructural, and socio-economic factors influencing the suitability of waste disposal sites.

The selected criteria include terrain characteristics, land use patterns, proximity to transportation networks, distance from settlements and sensitive areas, hydrological conditions, and groundwater depth. Each dataset was carefully chosen to ensure comprehensive spatial analysis and reliable decision-making.

The details of the datasets used in this study, along with their purpose and sources, are presented in Table 1.

Table 1. Spatial datasets used for landfill site suitability analysis

S. No.	Data Layer	Purpose	Source
1	DEM (Elevation)	Terrain analysis and slope evaluation	SRTM / ALOS
2	Land Use/Land Cover (LULC)	Identification of suitable land categories	USGS / ESA Satellite Data
3	Road Network	Accessibility and transportation analysis	OpenStreetMap (OSM)
4	Railway Network	Buffer restriction and safety consideration	OpenStreetMap (OSM)
5	Urban Settlements	Avoidance of densely populated areas	Satellite Data / OSM
6	Water Bodies	Environmental protection and buffer zones	Satellite Data / OSM
7	Groundwater Depth	Assessment of contamination risk	CGWB Reports
8	Sensitive Locations	Protection of schools, hospitals, etc.	Digitized from maps / OSM

Following data collection, all datasets were preprocessed and converted into a uniform coordinate system (UTM Zone 44N) to ensure spatial consistency and compatibility during analysis. The datasets were further clipped to the study area boundary and transformed into raster format for integration in the GIS-based multi-criteria evaluation process.

The use of open-source datasets ensures cost-effectiveness, transparency, and reproducibility, making the methodology suitable for both academic research and practical urban planning applications (Longley et al., 2015).

3.3 Data Preprocessing

Before performing spatial analysis, all collected datasets were subjected to a series of preprocessing steps to ensure accuracy, consistency, and compatibility within the GIS environment. Data preprocessing is a crucial step in GIS-based multi-criteria analysis, as inconsistencies in spatial reference, resolution, or format can lead to significant errors in the final results (Longley et al., 2015). The preprocessing steps adopted in this study are described below.

3.3.1 Projection

All spatial datasets were projected into a common coordinate reference system, specifically Universal Transverse Mercator (UTM) Zone 44N, to maintain spatial uniformity. Projection ensures that all datasets align correctly in geographic space and allows accurate distance and area calculations during analysis (Longley et al., 2015).

3.3.2 Clipping

The collected datasets were clipped using the administrative boundary of Gorakhpur City. This step restricted the analysis to the defined study area and eliminated unnecessary data outside the region of interest, thereby improving computational efficiency and accuracy.

3.3.3 Rasterization

Vector datasets such as road networks, railway lines, urban settlements, and sensitive locations were converted into raster format. Rasterization is essential for performing spatial operations such as overlay analysis and map algebra, which are fundamental components of GIS-based multi-criteria decision analysis (Malczewski, 2006).

3.3.4 Data Standardization

All raster layers were standardized to a common spatial resolution, extent, and coordinate system to ensure compatibility during overlay analysis. Standardization allows multiple thematic layers to be integrated accurately in the weighted overlay process.

3.4 Thematic Layer Preparation

In this stage, multiple thematic layers representing environmental, infrastructural, and socio-economic criteria were generated using GIS-based spatial analysis. These layers play a crucial role in determining the suitability of locations for landfill site selection. Each layer

was derived from processed datasets and represents a specific factor influencing environmental safety and operational feasibility.

The preparation of thematic layers involved buffer analysis, distance mapping, and spatial classification, which are essential components of GIS-based multi-criteria decision analysis (Malczewski, 2006; Nas et al., 2010).

3.4.1 Distance from Surface Water

Surface water bodies such as rivers, lakes, and ponds were identified and buffer zones were generated to prevent landfill development near water sources. This is essential to minimize the risk of leachate contamination entering surface water systems.

The distance from surface water was calculated using proximity analysis, and the resulting raster layer was classified into different suitability zones. The spatial distribution of distance from surface water is shown in Figure 3.

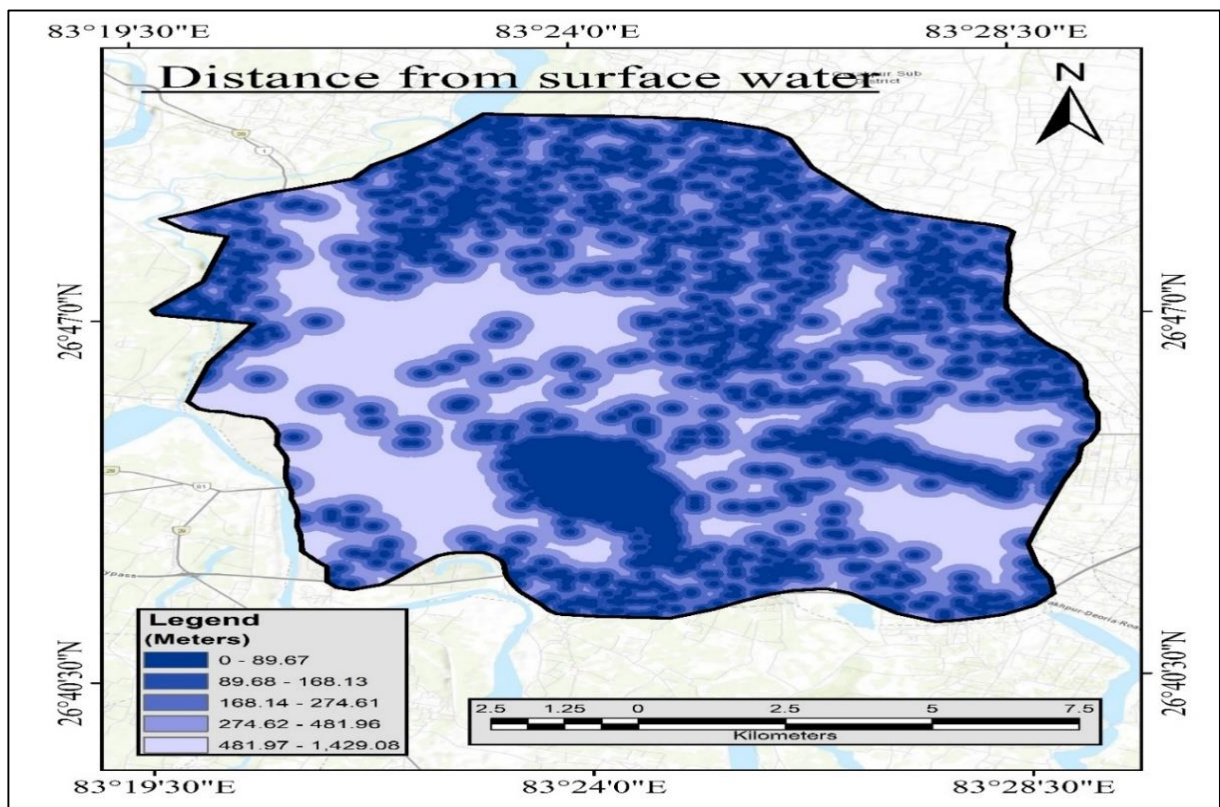


Figure 3. Distance from surface water bodies in Gorakhpur City showing proximity-based suitability zones.

3.4.2 Distance from Urban Areas

Urban settlement data was used to ensure that landfill sites are located at a safe distance from densely populated regions. Proximity to urban areas can lead to health hazards, odor issues, and social conflicts. Buffer zones were created around settlement areas, and distance analysis was performed to identify suitable zones. The distance from urban areas is illustrated in Figure 4.

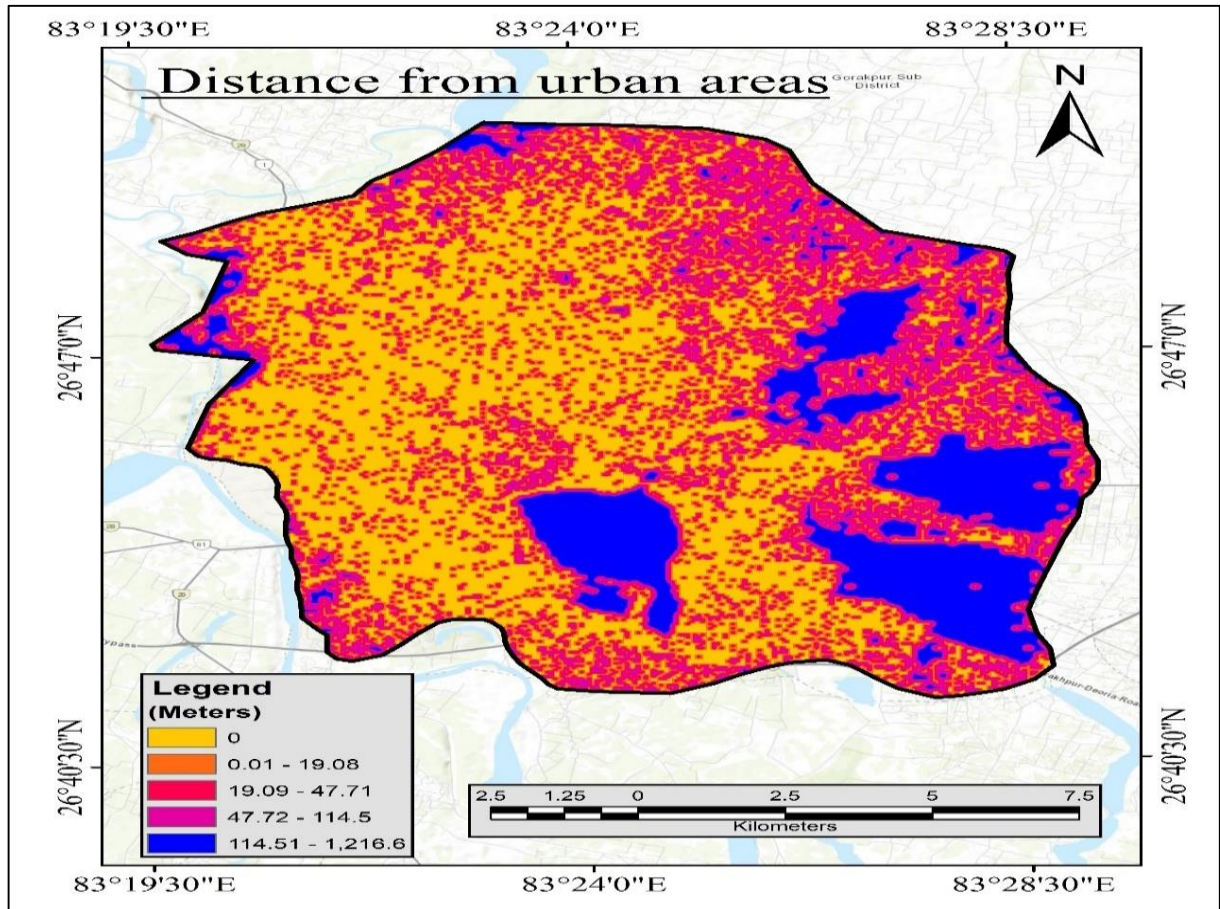


Figure 4. Distance from urban areas in Gorakhpur City showing proximity-based suitability zones.

3.4.3 Distance from Roads

The road network was analyzed to ensure accessibility for waste transportation while maintaining a suitable distance to avoid direct environmental and social disturbances. Distance analysis was performed to generate a road proximity map, which was further classified into suitability categories. The spatial distribution of distance from roads is

presented in Figure 5.

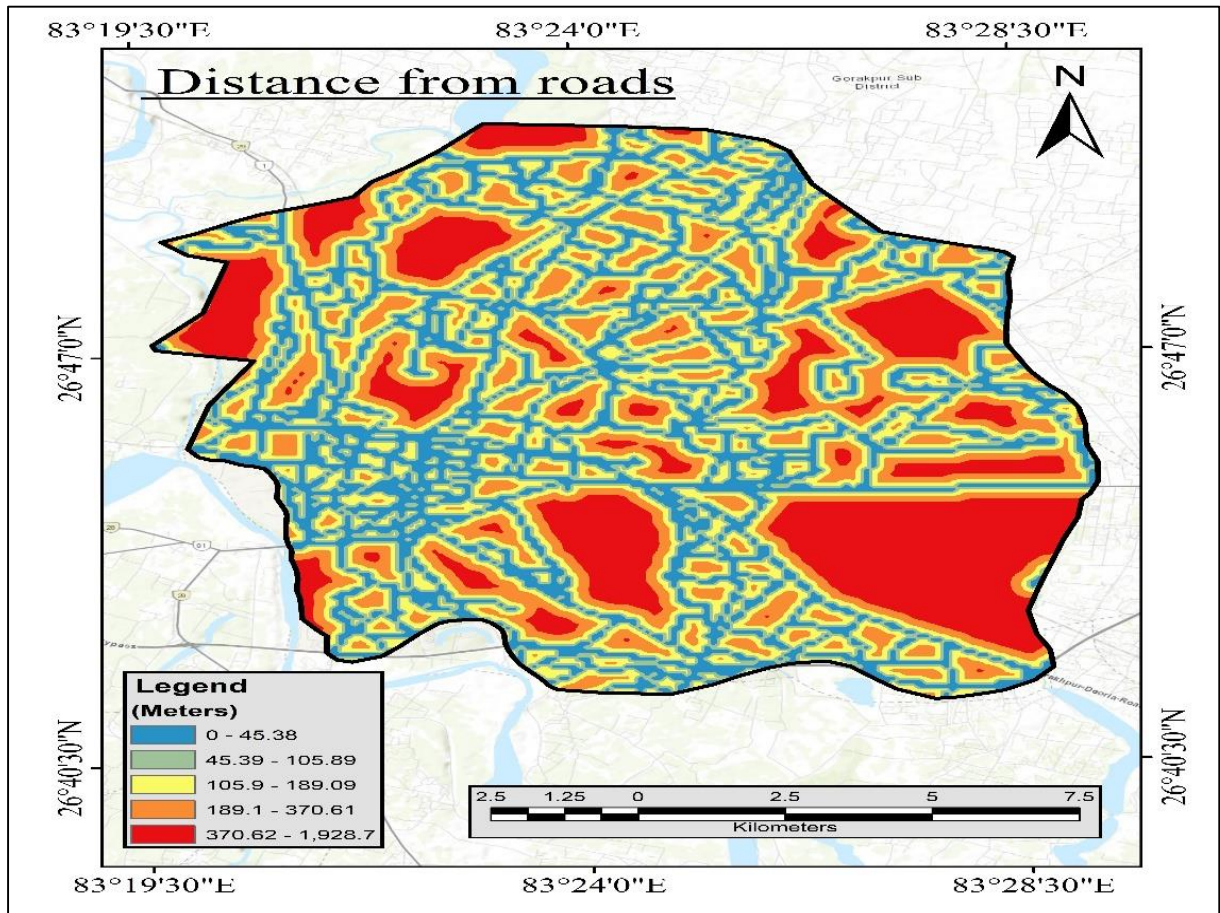


Figure 5. Distance from roads in Gorakhpur City showing proximity-based suitability zones.

3.4.4 Distance from Railways

Railway lines were considered as restricted zones to avoid potential hazards and land use conflicts. Buffer zones were created around railway networks, and distance analysis was performed. The distance from railway networks is shown in Figure 6.

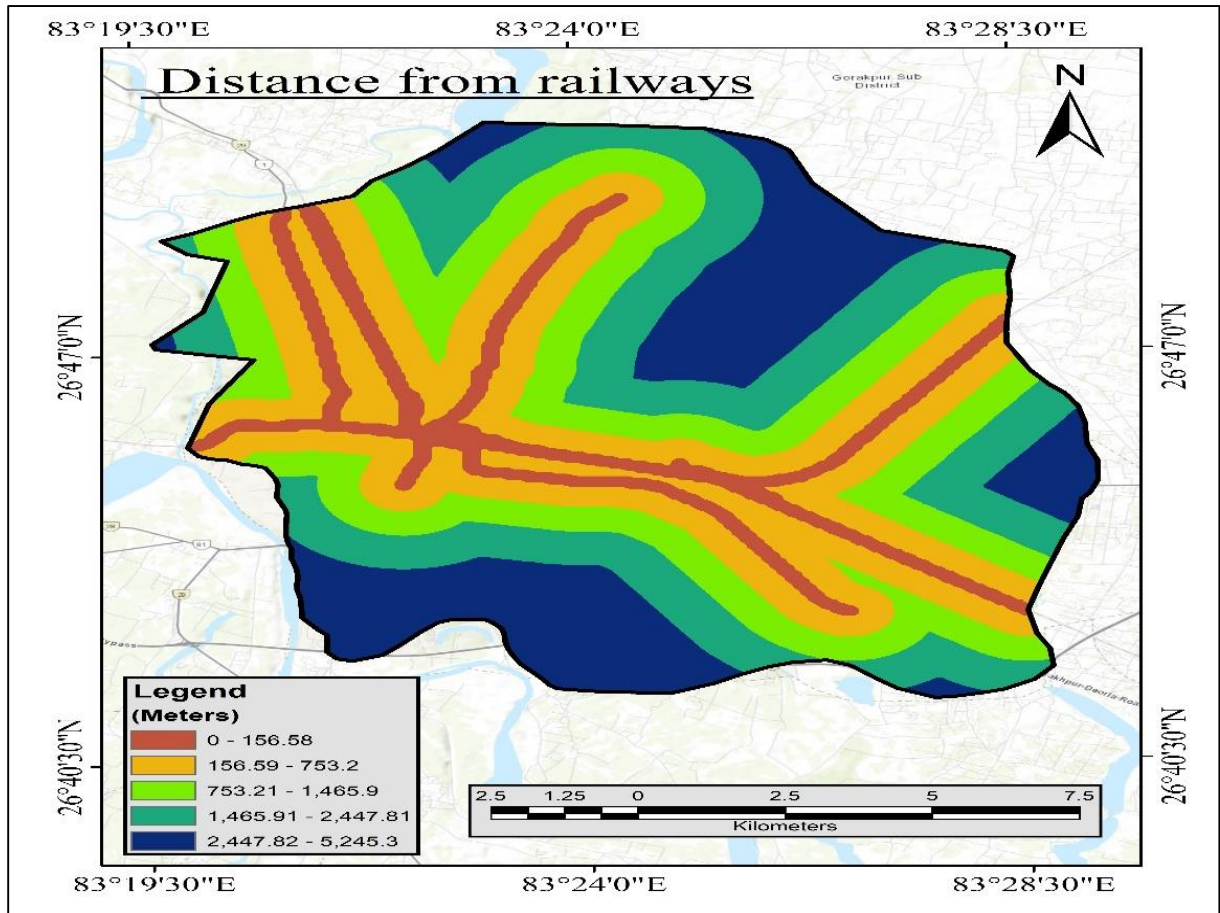


Figure 6. Distance from railways in Gorakhpur City showing buffer-based proximity zones for suitability analysis.

3.4.5 Distance from Sensitive Locations

Sensitive locations such as schools, hospitals, religious places, and protected areas were identified and buffered to ensure environmental safety and social acceptability. Distance analysis was conducted to avoid landfill placement near such locations. The spatial distribution of distance from sensitive locations is shown in Figure 7.

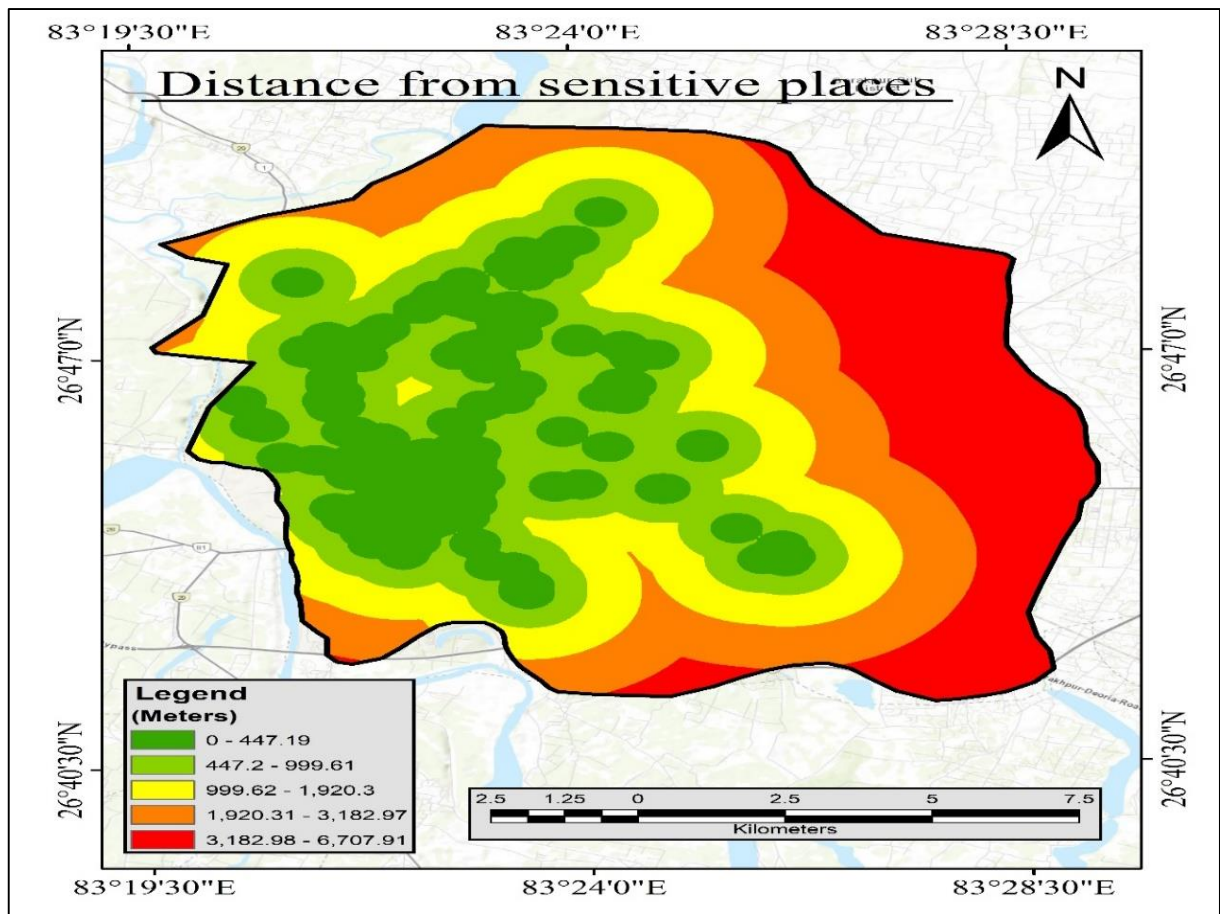


Figure 7. Distance from sensitive locations in Gorakhpur City showing buffer-based suitability zones for landfill site selection.

3.4.6 Groundwater Depth

Groundwater depth is a critical factor in landfill site selection, as shallow groundwater increases the risk of contamination due to leachate infiltration. Groundwater data was analyzed and classified into different suitability categories based on depth. The groundwater depth distribution is illustrated in Figure 8.

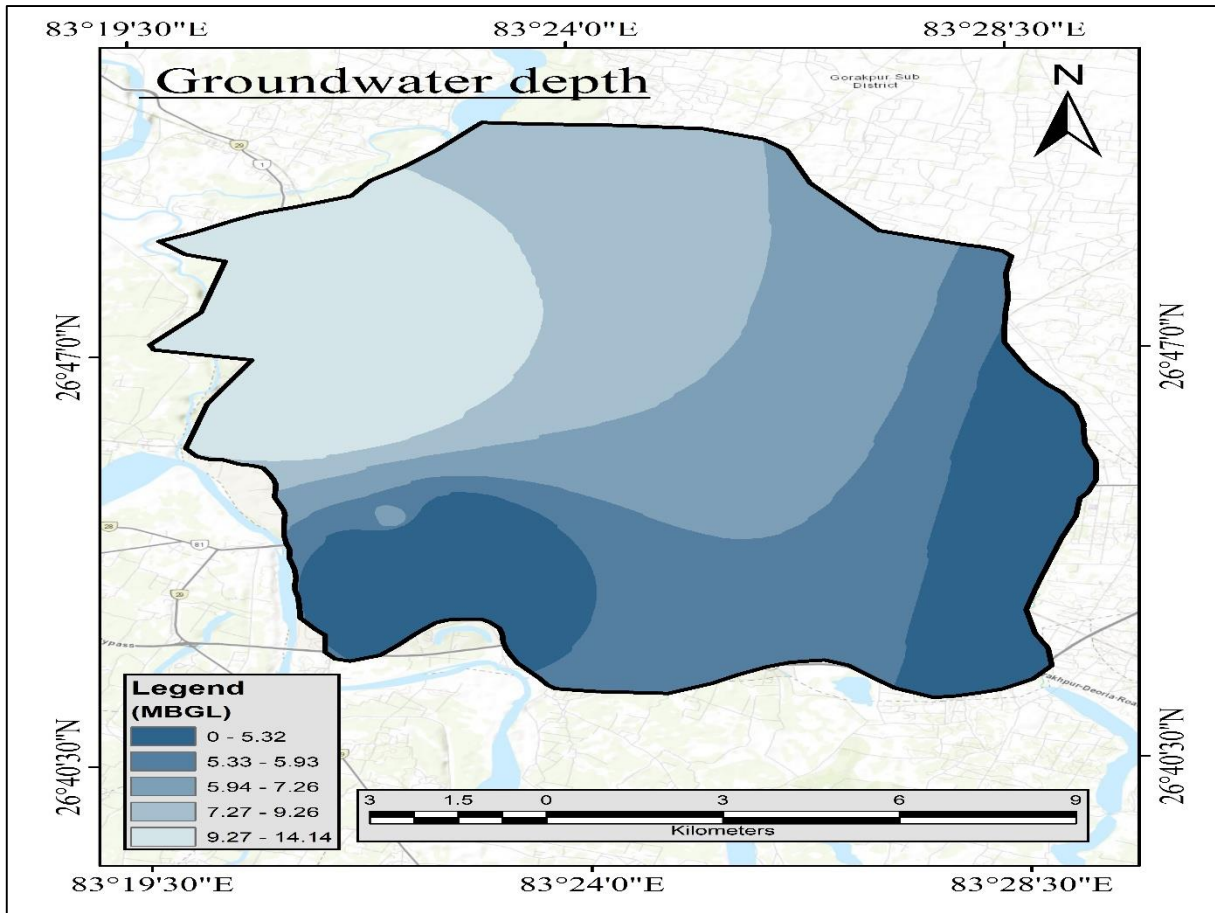


Figure 8. Spatial distribution of groundwater depth in Gorakhpur City indicating variation in groundwater levels for landfill suitability analysis.

3.4.7 Land Elevation

Elevation data derived from the Digital Elevation Model (DEM) was used to identify low-lying and flood-prone areas. Landfills are generally avoided in such areas to prevent waterlogging and contamination risks. The elevation layer was classified into suitable and unsuitable zones. The elevation map of the study area is presented in Figure 9.

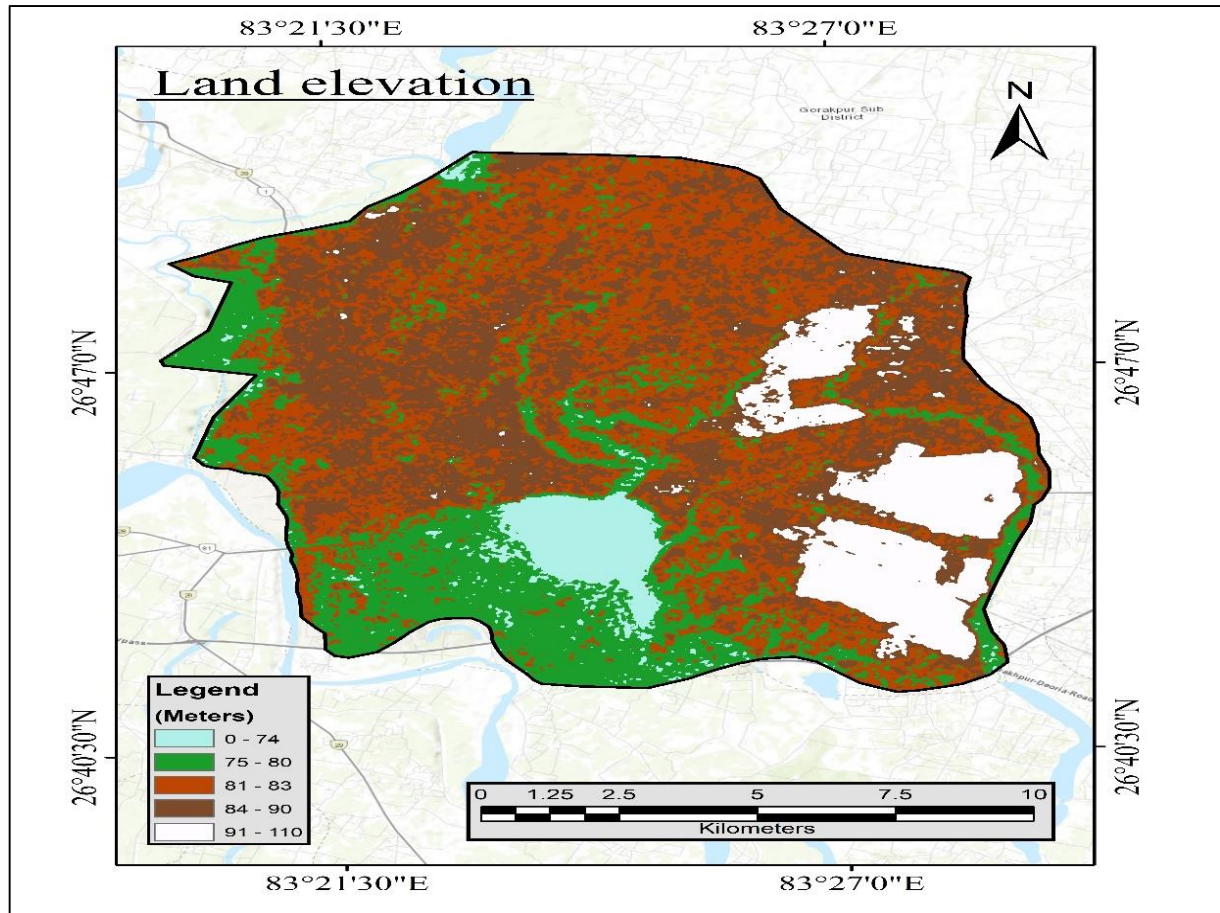


Figure 9. Spatial distribution of land elevation in Gorakhpur City derived from DEM, indicating elevation-based suitability for landfill site selection.

3.4.8 Land Use Land Cover (LULC)

Land Use Land Cover (LULC) analysis was carried out to identify suitable land categories for landfill development. Areas such as barren land and wasteland are generally considered suitable, while agricultural land, forests, and built-up areas are avoided. The LULC map was classified into different categories based on suitability criteria. The LULC distribution of the study area is shown in Figure 10.

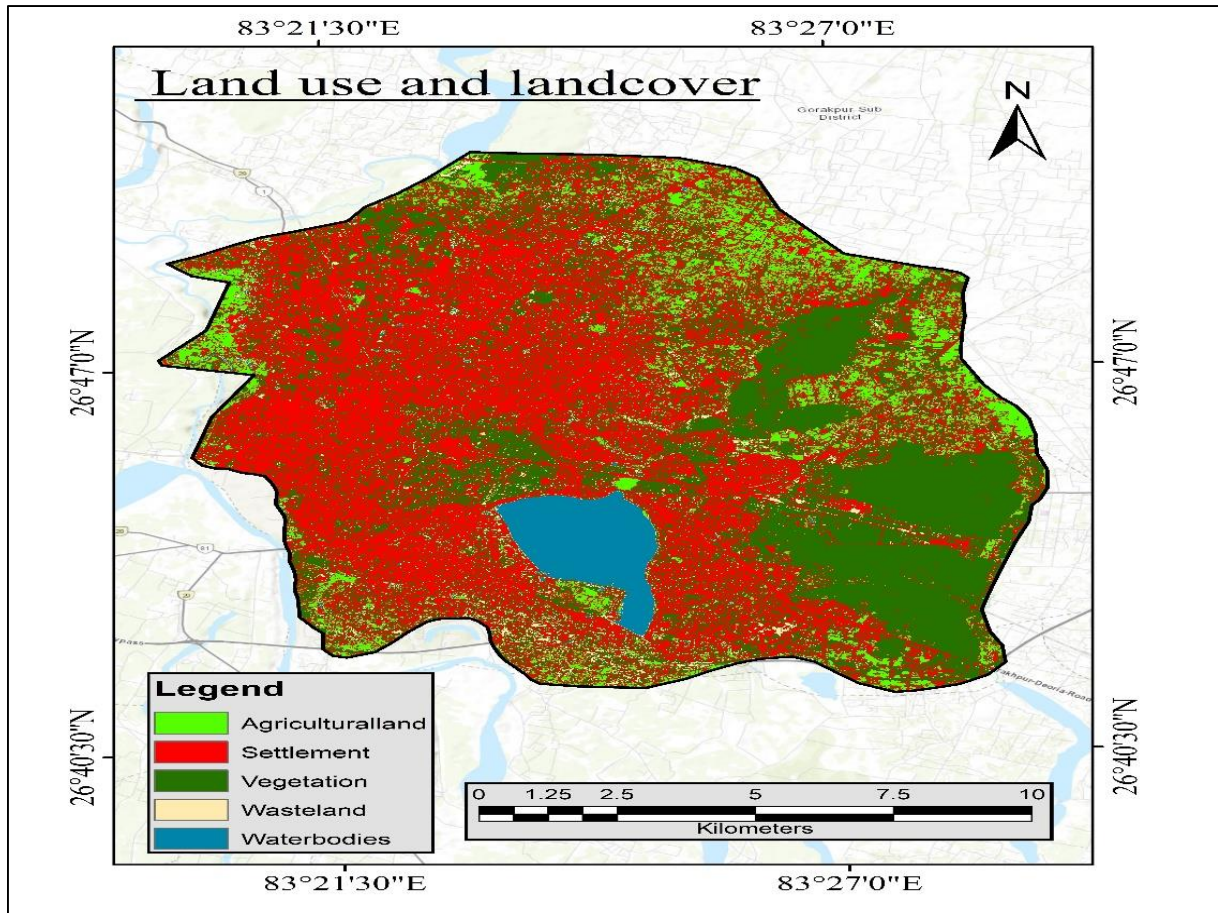


Figure 10. Land use and land cover (LULC) classification of Gorakhpur City showing major land categories for landfill suitability analysis.

3.5 Buffer Analysis

Buffer analysis was performed to create exclusion zones around environmentally sensitive and infrastructure-related features. Buffering is a fundamental GIS technique used to define zones of influence around spatial features and is widely applied in landfill site selection to ensure environmental safety (Malczewski, 2006; Nas et al., 2010).

In this study, buffer zones were generated around the following features:

- a) Water bodies
- b) Urban areas
- c) Railway networks
- d) Sensitive locations (schools, hospitals, religious places, etc.)

Different buffer distances were applied based on environmental guidelines and previous

research studies to minimize the potential risks associated with landfill development. Areas falling within restricted buffer zones were classified as unsuitable, as they may lead to groundwater contamination, public health issues, or land-use conflicts.

Buffer analysis was integrated with distance mapping techniques to generate proximity-based suitability layers, which were further used in the multi-criteria evaluation process (Kontos et al., 2005).

3.6 Reclassification of Criteria

To enable integration of multiple thematic layers, each criterion was reclassified into standardized suitability classes. Reclassification is essential in GIS-based multi-criteria analysis, as it converts diverse datasets into a common evaluation scale (Malczewski, 2006). Each thematic layer was categorized into the following suitability classes:

- a) Highly suitable
- b) Moderately suitable
- c) Less suitable
- d) Unsuitable

Numerical values (e.g., 1–4 or 1–5) were assigned to each class, where higher values represent higher suitability for landfill site selection. The classification thresholds were determined based on environmental standards, previous studies, and the relative importance of each parameter (Nas et al., 2010).

For example:

- a) Areas far from water bodies and settlements → Highly suitable
- b) Areas near sensitive features → Unsuitable
- c) Areas with deeper groundwater → More suitable

This process ensured uniformity across all layers, making them compatible for weighted overlay analysis.

3.7 Weighted Overlay Analysis

The weighted overlay technique was used to integrate all reclassified thematic layers into a single composite suitability map. This method is widely recognized for its effectiveness in spatial decision-making involving multiple criteria (Malczewski, 2006; Saaty, 1980).

In this approach, each criterion was assigned a weight based on its relative importance in

landfill site selection. Environmentally critical parameters were assigned higher weights, while secondary factors were given moderate weights.

For example:

- a) Distance from surface water bodies → High weight
- b) Groundwater depth → High weight
- c) Distance from urban areas → High weight
- d) Distance from roads → Moderate weight
- e) Land use/land cover → Moderate weight
- f) Distance from railways and sensitive locations → Moderate weight

The weighted overlay analysis was carried out using raster-based GIS operations, where each cell value represents the cumulative suitability score derived from all criteria.

The suitability index (SI) is expressed as Equation (1):

$$SI = \sum(W_i * X_i) \quad (1)$$

where:

W_i = weight of the i th criterion

X_i = reclassified score of the i th criterion

This approach provides a quantitative framework for evaluating spatial suitability and supports objective decision-making (Kontos et al., 2005; Nas et al., 2010).

3.8 Site Suitability Modeling Framework

The final stage of the analysis involved classifying the composite suitability map into distinct suitability zones. Based on the weighted overlay results, the study area was categorized into:

- a) Highly suitable zones
- b) Moderately suitable zones
- c) Less suitable zones
- d) Unsuitable zones

These zones represent the degree of suitability of different locations for landfill site development. Areas classified as highly suitable satisfy most of the environmental and infrastructural criteria, while unsuitable areas fail to meet essential requirements such as safe distance from water bodies, settlements, and sensitive locations.

4.1 Overview of Results

This chapter presents the detailed results obtained from the GIS-based multi-criteria decision analysis (MCDA) approach applied for landfill site selection in Gorakhpur City. The results derived from individual thematic layers are critically analyzed to understand their influence on landfill suitability.

Further, the integration of these layers through weighted overlay analysis provides a comprehensive spatial representation of suitability zones. The results highlight the combined impact of environmental, infrastructural, and socio-economic parameters on landfill site selection and provide a scientifically robust basis for decision-making.

4.2 Analysis of Thematic Layers

4.2.1 Distance from Surface Water

The distance from surface water (Figure 3) plays a critical role in determining landfill suitability due to the potential risk of leachate contamination. The analysis indicates that areas located within close proximity to rivers, lakes, and ponds are highly unsuitable, as contaminants from landfill sites can easily infiltrate surface water systems.

It is observed that central and low-lying regions of the study area, which are closer to water bodies, exhibit low suitability. In contrast, areas located at greater distances demonstrate higher suitability. This pattern suggests that maintaining adequate buffer distances from water bodies is essential for environmental protection and aligns with standard landfill siting guidelines.

4.2.2 Distance from Urban Areas

The distance from urban areas (Figure 4) reveals a clear relationship between population density and landfill suitability. Densely populated zones are identified as highly unsuitable due to potential health risks, odor nuisance, and social resistance.

The analysis shows that the central urban core of Gorakhpur exhibits low suitability, while peripheral regions demonstrate higher suitability. This indicates that landfill sites should be

strategically located away from residential zones to minimize human exposure and socio-environmental conflicts.

4.2.3 Distance from Roads

The road network analysis (Figure 5) highlights the importance of accessibility in landfill site selection. Areas located at moderate distances from roads are found to be most suitable, as they provide ease of transportation while minimizing direct environmental and social disturbances.

Areas too close to major roads may lead to pollution and traffic issues, whereas areas too far increase transportation costs. Therefore, an optimal distance range is necessary to balance operational efficiency and environmental considerations.

4.2.4 Distance from Railways

The railway proximity analysis (Figure 6) indicates that areas near railway lines are unsuitable due to safety risks and regulatory constraints. Buffer zones around railway infrastructure effectively exclude these areas from landfill consideration.

Suitable areas are those located beyond the restricted zones, ensuring minimal interference with transportation infrastructure.

4.2.5 Distance from Sensitive Locations

The analysis of sensitive locations (Figure 7) shows that proximity to schools, hospitals, and other socially important sites significantly reduces suitability. These areas require strict protection due to their importance for public health and well-being.

The results indicate that regions located at safe distances from sensitive locations are more suitable, emphasizing the need for careful spatial planning in landfill siting.

4.2.6 Groundwater Depth

Groundwater depth (Figure 8) is a crucial parameter influencing landfill suitability. The analysis indicates that areas with deeper groundwater levels are more suitable, as they reduce the risk of contamination from leachate infiltration.

Conversely, areas with shallow groundwater levels are highly vulnerable and therefore unsuitable. This highlights the importance of hydrogeological considerations in landfill site selection.

4.2.7 Land Elevation

The elevation analysis (Figure 9) reveals that low-lying areas are prone to flooding and

waterlogging, making them unsuitable for landfill development. Flood-prone zones increase the risk of leachate spread and environmental contamination.

Higher elevation areas exhibit better drainage conditions and are therefore more suitable. This indicates that topography plays a significant role in determining landfill suitability.

4.2.8 Land Use Land Cover (LULC)

The LULC analysis (Figure 10) demonstrates that barren and unused lands are the most suitable for landfill development, as they minimize land-use conflicts and environmental impact.

Agricultural land, built-up areas, and vegetated regions are identified as unsuitable due to their economic and ecological importance. This highlights the need to prioritize non-productive land for waste disposal.

4.3 Weighted Overlay Results

The weighted overlay analysis represents the integration of all reclassified thematic layers into a single composite suitability map, providing a holistic evaluation of landfill site suitability. This approach enables the simultaneous consideration of multiple environmental, infrastructural, and socio-economic factors, each assigned a specific weight based on its relative importance (Malczewski, 2006; Saaty, 1980). The weights assigned to each criterion used in the analysis are presented in Table 2, while the resulting spatial distribution of landfill suitability zones is illustrated in Figure 11.

The results indicate that environmental parameters, particularly distance from surface water bodies and groundwater depth, exert the strongest influence on landfill suitability. Areas located far from water bodies and characterized by deeper groundwater levels consistently exhibit higher suitability scores. This highlights the critical role of hydrological factors in minimizing environmental risks associated with landfill operations.

In addition, distance from urban areas and sensitive locations significantly contributes to the suitability classification. Regions located away from densely populated areas and socially sensitive zones demonstrate higher suitability due to reduced health risks and improved social acceptability.

The integration of road network proximity ensures that selected areas maintain adequate accessibility for waste transportation, while avoiding excessive environmental disturbance.

Similarly, the inclusion of land use/land cover (LULC) data ensures that ecologically and economically valuable land types are excluded from consideration.

The composite suitability map reveals that areas satisfying multiple favorable criteria simultaneously are classified as highly suitable, whereas areas failing to meet key environmental thresholds are categorized as unsuitable. This demonstrates the effectiveness of the GIS-based MCDA approach in synthesizing complex spatial information into a single decision-support framework.

Table 2. Criteria Weights Used in Weighted Overlay Analysis

S. No.	Criteria	Weight (%)	Justification
1	Distance from Surface Water	20	Critical to prevent water contamination
2	Groundwater Depth	18	Reduces risk of leachate infiltration
3	Distance from Urban Areas	15	Minimizes health hazards and social conflicts
4	Distance from Sensitive Locations	12	Ensures safety of public institutions
5	Land Use/Land Cover (LULC)	12	Identifies suitable land types
6	Distance from Roads	10	Ensures accessibility for waste transport
7	Distance from Railways	7	Maintains safety from transport infrastructure
8	Land Elevation	6	Avoids flood-prone and low-lying areas

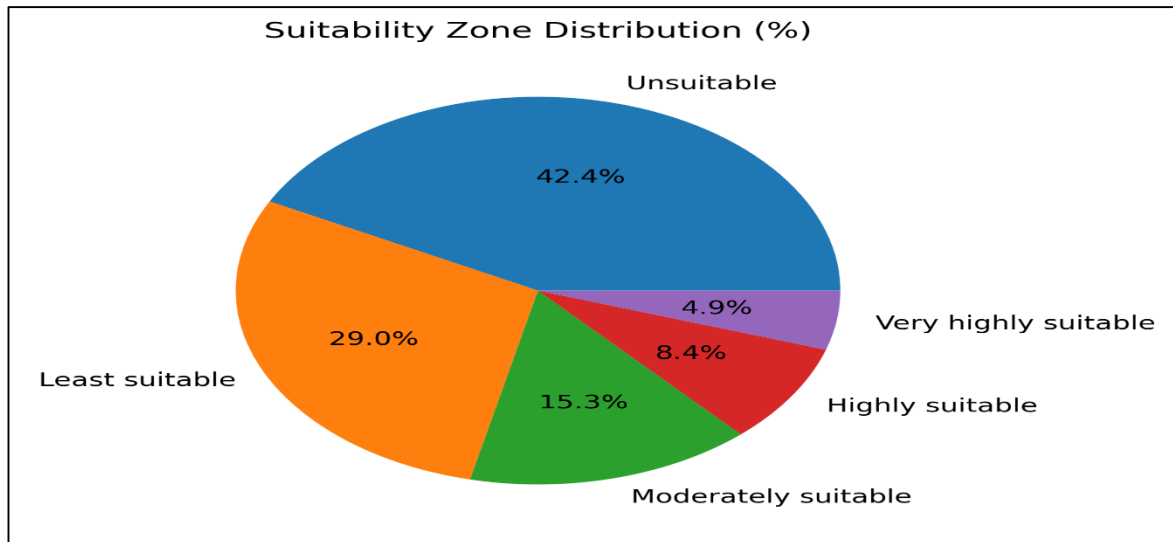


Figure 11. Final landfill site suitability map of Gorakhpur City derived from GIS-based weighted overlay analysis, showing spatial distribution of unsuitable, least suitable, moderately suitable, highly suitable, and very highly suitable zones

4.4 Site Suitability Analysis

The final landfill suitability map (Figure 12) provides a comprehensive spatial representation of suitability zones across Gorakhpur City. The classification of the study area into different suitability categories reflects the cumulative influence of all thematic layers integrated through the weighted overlay process.

The highly suitable zones are predominantly located in the peripheral regions of the study area. These areas are characterized by:

- a) Low population density
- b) Greater distance from surface water bodies
- c) Deeper groundwater levels
- d) Suitable land use types such as barren or unused land
- e) Moderate accessibility to road networks

These conditions collectively ensure minimal environmental impact and operational feasibility, making these zones ideal for landfill development.

The moderately suitable zones are distributed across transitional regions, where some criteria are satisfied while others present moderate limitations. These areas may require additional planning and mitigation measures before being considered for landfill siting.

The less suitable zones are generally located near semi-urban areas or regions with partial environmental constraints. These areas may exhibit proximity to settlements, moderate groundwater depth, or less favorable land use characteristics.

The unsuitable zones are concentrated around:

- a) Water bodies
- b) Dense urban settlements
- c) Sensitive locations such as schools and hospitals
- d) Low-lying or flood-prone areas

These zones are excluded from landfill consideration due to high environmental risk and potential social conflicts.

The observed spatial pattern clearly reflects the combined influence of all criteria and validates the effectiveness of the GIS-based MCDA methodology. The results demonstrate that landfill suitability is not determined by a single factor but by the interaction of multiple environmental and infrastructural parameters. The spatial distribution of these suitability

zones is illustrated in Figure 12.

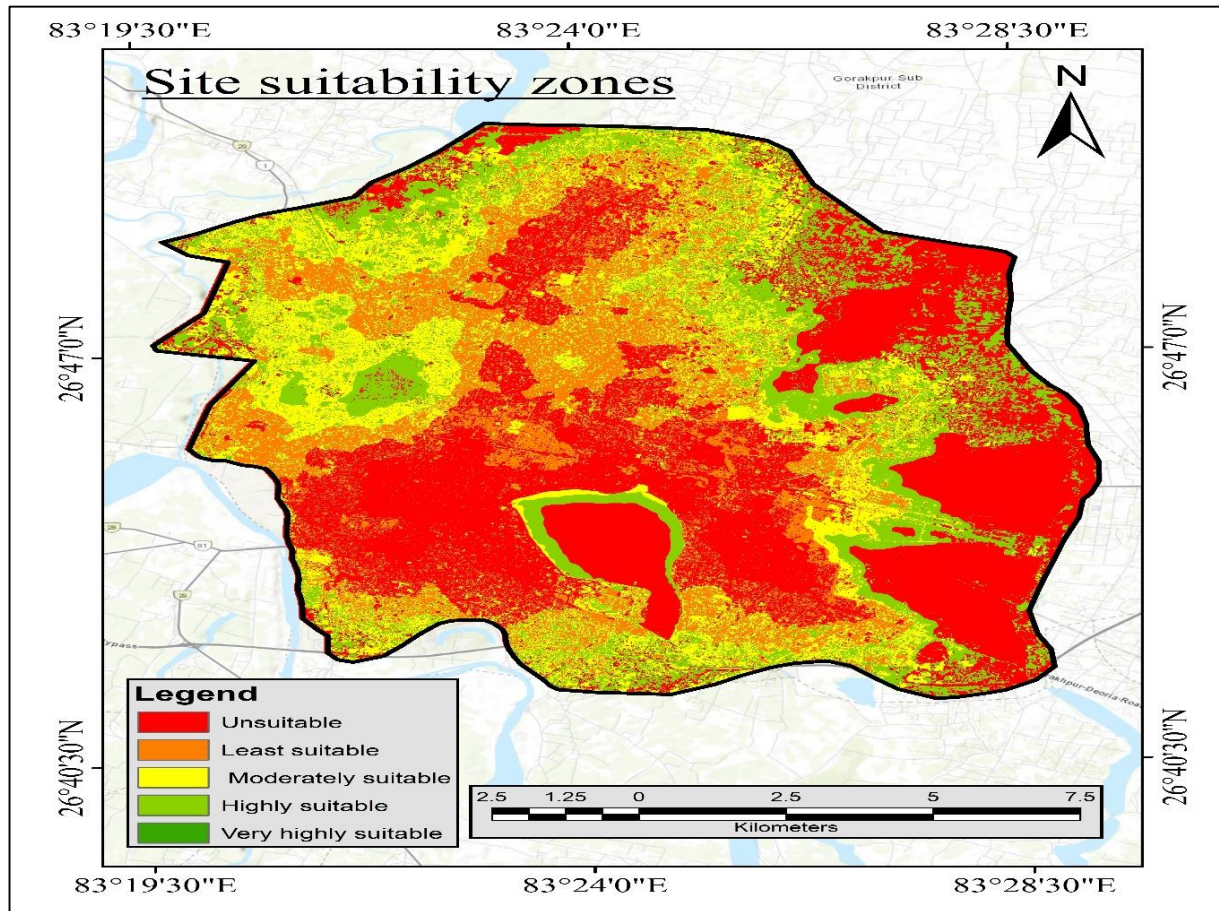


Figure 12. Final landfill site suitability map of Gorakhpur City derived from GIS-based weighted overlay analysis, showing zones classified as unsuitable, least suitable, moderately suitable, highly suitable, and very highly suitable.

4.5 Identification of Optimal Sites

Based on the integrated analysis of all thematic layers and the results of the weighted overlay, the optimal landfill sites were identified within the study area. These locations correspond to zones classified as highly suitable, where the combined influence of environmental, infrastructural, and socio-economic criteria is most favorable.

The analysis indicates that the most suitable landfill sites are predominantly located in the peripheral regions of Gorakhpur City, particularly in areas that are relatively distant from

dense urban settlements and environmentally sensitive zones. These locations exhibit minimal risk of groundwater and surface water contamination, as they are situated at safe distances from rivers, lakes, and other water bodies.

In addition, the identified sites are characterized by deeper groundwater levels, which further reduces the potential for leachate infiltration and contamination. From a topographical perspective, these areas generally lie in moderately elevated zones, which helps prevent issues related to flooding and waterlogging.

The selected locations also demonstrate favorable land use characteristics, primarily consisting of barren or unused land. This minimizes conflicts with existing land uses such as agriculture, residential areas, and ecological zones, thereby enhancing the feasibility of landfill development.

From an infrastructural standpoint, the optimal sites maintain a moderate proximity to road networks, ensuring efficient transportation of solid waste without causing excessive environmental disturbance or traffic congestion. At the same time, these areas are sufficiently distant from railway lines and sensitive locations such as schools and hospitals, ensuring compliance with safety and environmental guidelines.

Overall, the identified landfill sites represent a balanced combination of environmental safety, socio-economic acceptability, and operational feasibility. These locations are therefore considered highly suitable for sustainable solid waste management in Gorakhpur City.

The spatial distribution of these optimal zones corresponds to the high suitability areas depicted in Figure 12, which provides a clear visual representation of potential landfill locations.

4.6 Discussion

The results obtained from the present study clearly demonstrate that the integration of Geographic Information Systems (GIS) with Multi-Criteria Decision Analysis (MCDA) provides a robust and reliable framework for landfill site selection. The ability of GIS to handle spatial datasets and MCDA to assign relative importance to multiple criteria ensures a comprehensive and scientifically sound evaluation of environmental suitability (Malczewski, 2006; Saaty, 1980).

The analysis reveals that environmental factors such as distance from surface water bodies and groundwater depth play a dominant role in determining landfill suitability. Areas located far from water bodies and with deeper groundwater levels were identified as highly suitable, as they significantly reduce the risk of leachate contamination. This finding is consistent with earlier studies conducted by Kontos et al. (2005) and Nas et al. (2010), which emphasized the importance of hydrological parameters in landfill siting.

In addition to environmental criteria, proximity to urban settlements and sensitive locations was found to be a critical factor influencing site suitability. The results indicate that areas near densely populated regions, schools, and hospitals are unsuitable due to potential health hazards, odor nuisance, and socio-environmental conflicts. This highlights the importance of incorporating socio-economic considerations into landfill planning to ensure public acceptance and minimize negative impacts.

The role of infrastructural factors, particularly road accessibility, was also evident in the analysis. Areas with moderate proximity to road networks were found to be most suitable, as they facilitate efficient transportation of waste while avoiding excessive environmental disturbance. This balance between accessibility and environmental protection is essential for the operational feasibility of landfill sites.

Furthermore, the integration of land use/land cover (LULC) data into the analysis ensured that environmentally sensitive and economically valuable land types such as agricultural land and built-up areas were excluded from landfill consideration. The preference for barren and unused land highlights the importance of sustainable land-use planning in waste management.

The weighted overlay analysis effectively combined all thematic layers into a single composite suitability map, demonstrating the strength of the GIS-MCDA approach in handling complex spatial decision-making problems. The resulting suitability zones reflect the cumulative influence of all criteria and provide a clear basis for identifying optimal landfill locations.

Another significant aspect of this study is the use of open-source GIS platforms, which enhances the accessibility and reproducibility of the methodology. Unlike proprietary software, open-source tools provide cost-effective solutions for spatial analysis, making this approach highly suitable for implementation in developing regions where financial and

technical resources may be limited (Longley et al., 2015).

Overall, the findings of this study confirm that GIS-based MCDA is not only an effective tool for landfill site selection but also a scalable and adaptable framework that can be applied to other regions facing similar waste management challenges. The methodology provides a scientific basis for decision-making and can support urban planners and policymakers in developing environmentally sustainable waste management strategies.

CHAPTER 05

CONCLUSION AND RECOMMENDATIONS

5.1 Conclusion

The present study successfully applied a Geographic Information System (GIS)-based Multi-Criteria Decision Analysis (MCDA) approach to identify optimal locations for solid waste disposal in Gorakhpur City. By integrating multiple spatial datasets, including land use/land cover, groundwater depth, road networks, urban settlements, surface water bodies, and sensitive locations, a comprehensive framework for landfill site suitability assessment was developed.

The analysis demonstrated that landfill suitability is influenced by a combination of environmental, infrastructural, and socio-economic factors. Among these, distance from surface water bodies and groundwater depth emerged as the most critical parameters, as they directly affect the risk of environmental contamination. Areas located far from water bodies and characterized by deeper groundwater levels were identified as highly suitable for landfill development.

The results further revealed that peripheral regions of the study area exhibit the highest suitability due to lower population density, reduced environmental sensitivity, and better land availability. In contrast, areas near urban settlements, water bodies, and sensitive locations were found to be unsuitable due to potential environmental and public health risks.

The use of weighted overlay analysis enabled the integration of all criteria into a single composite suitability map, providing a clear and scientifically justified basis for decision-making. The final suitability map effectively delineated zones into highly suitable, moderately suitable, less suitable, and unsuitable categories, thereby facilitating the identification of optimal landfill locations.

Overall, the study confirms that the GIS-based MCDA approach is an effective, reliable, and practical tool for landfill site selection. The methodology ensures a systematic evaluation of multiple criteria and supports environmentally sustainable waste management planning. Furthermore, the use of open-source GIS tools enhances the accessibility and applicability of this approach for real-world implementation, particularly in developing regions.

5.2Key Findings

The major findings of the present study are summarized as follows:

- a) The integration of GIS and Multi-Criteria Decision Analysis (MCDA) proved to be an effective approach for identifying suitable landfill sites.
- b) Distance from surface water bodies was identified as the most critical factor influencing landfill suitability, as areas close to water bodies pose a high risk of contamination.
- c) Groundwater depth significantly influenced site selection, with deeper groundwater areas being more suitable due to reduced vulnerability to leachate infiltration.
- d) Proximity to urban settlements played a key role, as areas near densely populated regions were found to be unsuitable due to potential health hazards and social conflicts.
- e) Sensitive locations such as schools, hospitals, and religious sites must be avoided, highlighting the importance of socio-environmental considerations in landfill planning.
- f) Road network accessibility was found to be essential for operational feasibility, with areas at moderate distances from roads being the most suitable.
- g) Land Use/Land Cover (LULC) analysis indicated that barren and unused lands are the most appropriate for landfill development, while agricultural and built-up areas are unsuitable.
- h) Low-lying and flood-prone areas were identified as unsuitable due to increased risk of leachate spread and environmental contamination.
- i) The weighted overlay analysis successfully integrated all criteria and generated a composite suitability map, enabling the classification of the study area into four categories: highly suitable, moderately suitable, less suitable, and unsuitable.
- j) The peripheral regions of Gorakhpur City were identified as the most suitable zones for landfill site development due to favorable environmental and infrastructural conditions.

5.3 Recommendations

Based on the results of the present study, the following recommendations are proposed for

effective and sustainable solid waste management planning in Gorakhpur City:

- a) The highly suitable zones identified in the study should be prioritized by municipal authorities for the development of sanitary landfill sites, as these areas satisfy the required environmental and infrastructural criteria.
- b) Landfill development should be strictly avoided in flood-prone and low-lying areas, as these regions increase the risk of leachate dispersion and environmental contamination, particularly during monsoon seasons.
- c) Adequate buffer distances must be maintained from critical features such as surface water bodies, urban settlements, railway lines, and sensitive locations (schools, hospitals, etc.) to minimize environmental and public health risks.
- d) Groundwater conditions should be carefully evaluated before site selection, and areas with shallow groundwater levels should be excluded to prevent contamination.
- e) Preference should be given to barren or unused land for landfill development in order to minimize land-use conflicts and preserve agricultural and ecologically sensitive areas.
- f) Proper road connectivity should be ensured for selected landfill sites to facilitate efficient waste transportation while avoiding excessive disturbance to nearby communities.
- g) Municipal authorities should adopt GIS-based decision-support systems for future urban planning and waste management, as it allows systematic and data-driven site selection.
- h) The use of open-source GIS platforms is recommended to reduce implementation costs and ensure accessibility for local planning agencies.
- i) Periodic monitoring and environmental impact assessment should be conducted for selected landfill sites to ensure long-term sustainability and compliance with environmental regulations.

5.4 Limitations of the Study

While the present study provides a systematic and scientific framework for landfill site selection using GIS-based MCDA, certain limitations should be acknowledged:

- a) The analysis is primarily based on secondary data sources, which may contain

inherent inaccuracies or may not fully represent current ground conditions.

- b) The final landfill suitability map (Figure 11) provides a comprehensive spatial representation of suitability zones across Gorakhpur City. The classification of the study area into different suitability categories reflects the cumulative influence of all thematic layers integrated through the weighted overlay process.
- c) The spatial resolution of some datasets, particularly DEM and LULC, may limit the precision of the analysis and affect the accuracy of suitability classification.
- d) The assignment of weights to different criteria was based on general knowledge and literature review, which may introduce a degree of subjectivity in the decision-making process.
- e) Certain important factors such as soil characteristics, land ownership, economic feasibility, and social acceptance were not explicitly considered due to data limitations.

Despite these limitations, the study provides a reliable preliminary framework for landfill site selection and can be further refined with detailed field investigations and higher-resolution data.

5.5 Future Scope

The present study can be further enhanced and extended in several ways to improve the accuracy and applicability of landfill site selection:

- a) The use of advanced multi-criteria techniques such as the Analytical Hierarchy Process (AHP) can be incorporated to assign more accurate and scientifically justified weights to different criteria.
- b) High-resolution spatial datasets, including detailed DEM, LULC, and groundwater data, can be used to improve the precision of the analysis.
- c) Field surveys and ground validation should be conducted to verify the suitability of the identified sites and ensure practical feasibility.
- d) Additional parameters such as soil properties, land ownership, economic cost analysis, and social acceptance can be integrated into the analysis for more comprehensive decision-making.
- e) The application of machine learning and artificial intelligence techniques can

further enhance the accuracy and efficiency of landfill site suitability modeling.

- f) The developed methodology can be extended to other cities and regions, providing a scalable framework for sustainable solid waste management planning.

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PUBLICATIONS AND PRESENTATIONS

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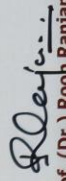
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Certificate of Presentation

This is to certify that Ms./Mr./Dr. Adarsh Pal Year: 4th Year

Branch: Civil Engineering has presenting the paper/abstract titled: Open - Source GIS Approach for Identifying Optimal Solid Waste Management sites in Gorakhpur

the "National Conference on Sustainable Civil Engineering & Geospatial Technologies: Materials & Innovations for Climate Resilience (NCRT-SCE 2026)" held from 15th to 16th April, 2026.

We congratulate the participant for outstanding performance and contribution.

 Mr. Pratish Kannaujia Organizing Secretary	 Mr. Ankur Kumar Joint Organizing Secretary	 Mr. Baijnath Nishad Conference Convener	 Prof. (Dr.) Roop Ranjan Conference Chair
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